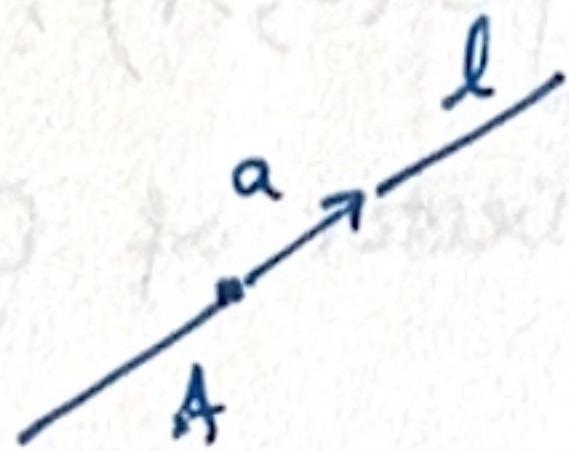


2.1. Parametric equations for the line l .a) l contains $A(1,2)$ and l is $\parallel$ to the vector $a(3,-1)$.

$$l: \begin{cases} x = 1 + 3t \\ y = 2 - t \end{cases}, t \in \mathbb{R}$$



$$\Rightarrow l: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} t, t \in \mathbb{R}.$$

$$\Rightarrow l: \begin{cases} x = 1 + 3t \\ y = 2 - t \end{cases}, t \in \mathbb{R}.$$

b) l contains the origin and is $\parallel$ to $b(4,5)$.

$$l: \begin{cases} x = 0 + 4s \\ y = 0 + 5s \end{cases}, s \in \mathbb{R}$$

c) l contains $K(1,7)$ and is $\parallel O_y$. $\parallel O_y \Rightarrow l$ has the same direction vectors as O_y The basis of $\mathbb{R}^2$ is $(i, j) \Rightarrow j$ is a direction vector for $l \Rightarrow$

$$\Rightarrow \text{the components of } j \text{ are } (0,1). \Rightarrow l: \begin{cases} x = 1 \\ y = 7 + t \end{cases}, t \in \mathbb{R}$$

2.2. For the lines in 2.1,

a) det. a Cartesian eq. for l .

$$l: \begin{cases} x = 1 + 3t \\ y = 2 - t \end{cases}, t \in \mathbb{R} \Rightarrow \begin{cases} t = \frac{x-1}{3} \\ t = \frac{y-2}{-1} \end{cases} \Rightarrow l: \frac{x-1}{3} = \frac{y-2}{-1}$$

$$\Rightarrow l: -x + 1 = 3y - 6$$

$$l: x + 3y - 7 = 0$$

Since $a(3,-1)$ direction vector, all other direction vectors are:

$$\{ \lambda \cdot a = (3\lambda, -\lambda) : \lambda \in \mathbb{R} \setminus \{0\} \}.$$

b) describe all direction vectors for l .

$$l: \begin{cases} x = 4s \\ y = 5s \end{cases}, s \in \mathbb{R} \Rightarrow \begin{cases} s = \frac{x}{4} \\ s = \frac{y}{5} \end{cases} \Rightarrow l: \frac{x}{4} = \frac{y}{5} \Rightarrow l: 5x - 4y = 0.$$

Direction vectors: $\{ \lambda \cdot b(4,5) = (4\lambda, 5\lambda), \lambda \in \mathbb{R} \}$

e) $l: \begin{cases} x=1 \\ y=7+t \end{cases}$, because $(x,y) = (1,7) + t \cdot (0,1) = (1,7) + (0,t) = (1,7+t)$.

$\Rightarrow l: x=1$ (y arbitrary)

$\Rightarrow$ direction vectors: $\{ \lambda \cdot j = (0, \lambda) : \lambda \in \mathbb{R}^* \}$.

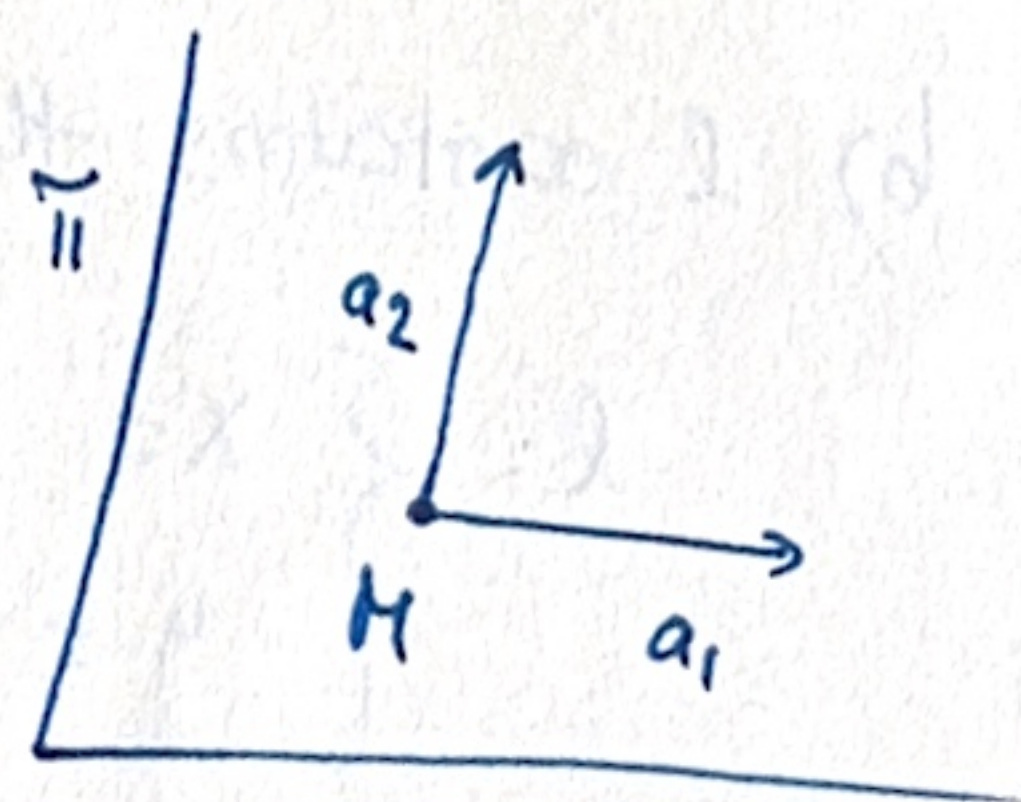
(because the direction vector of Oy is $(0,1)$).

2.3. Parametric eq. for plane π :

a) π contains $M(1,0,2)$ and is $\parallel$ to $a_1(3,-1,1)$ and $a_2(0,3,1)$.

$\pi: \begin{cases} x = 1 + 3t + 0 \cdot s \\ y = 0 - 1 \cdot t + 3 \cdot s \\ z = 2 + 1 \cdot t + 1 \cdot s \end{cases}, t, s \in \mathbb{R}.$

$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} \cdot t + \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \cdot s, t, s \in \mathbb{R}.$



$\Rightarrow \pi: x = 1 + 3t, y = -t + 3s, z = 2 + t + s, t, s \in \mathbb{R}.$

b) π contains $A(-2,1,1), B(0,2,3), C(1,0,-1)$.

$\vec{AB} = B(0,2,3) - A(-2,1,1) = (2,1,2) \Rightarrow \vec{AB} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

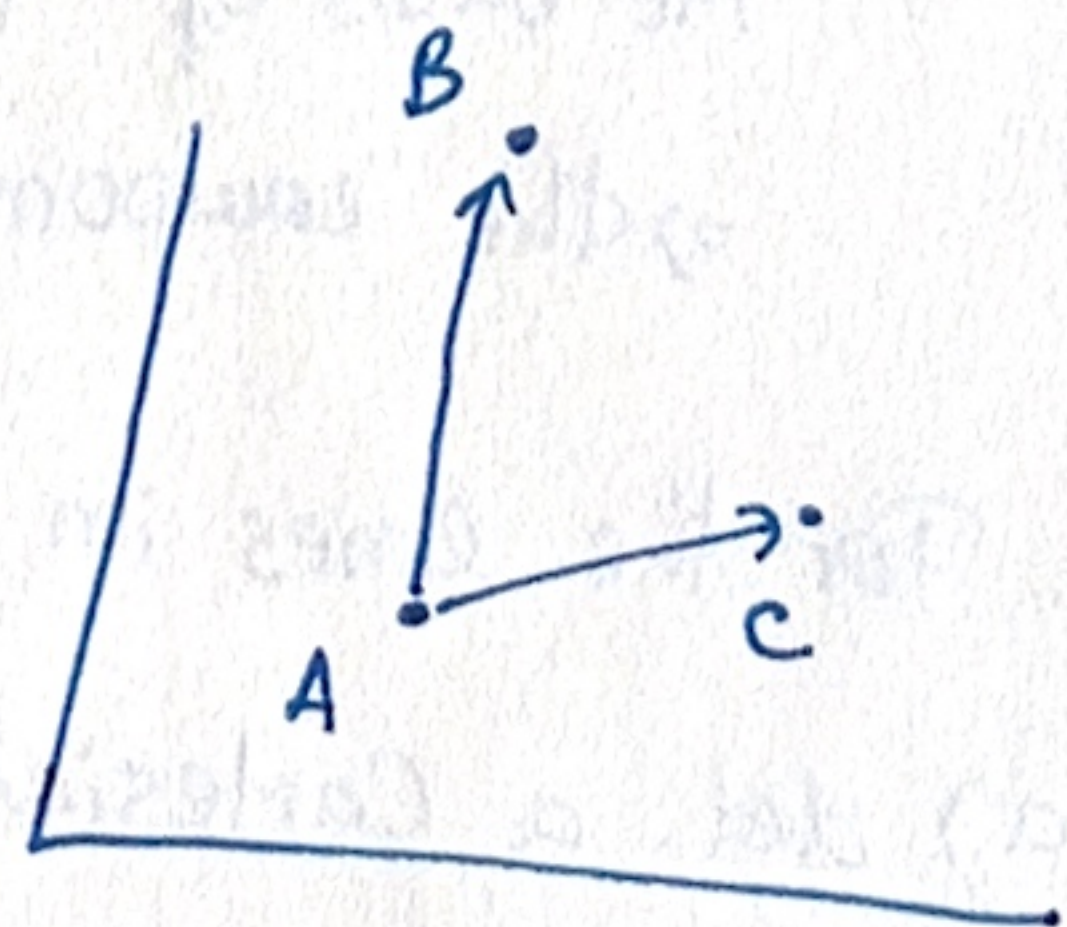
$\vec{AC} = C - A = (3, -1, -2) \Rightarrow \vec{AC} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$

$\vec{AB}, \vec{AC}$ are vectors represented with points in π .

$\vec{AB}, \vec{AC}$ lin. indep. $\Rightarrow$ they form a basis of the direction space $D(\pi)$.

$\pi: \begin{cases} x = -2 + 2t + 3s \\ y = 1 + t - s \\ z = 1 + 2t - 2s \end{cases}, t, s \in \mathbb{R}$

$\downarrow$ $\downarrow$ $\downarrow$
 $P.A.$ $\vec{AB}$ $\vec{AC}$
 (fixed)



$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \cdot t + \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} \cdot s, t, s \in \mathbb{R}.$

c) π contains the point $A(1,2,1)$ and is $\parallel$ to i and j .

$i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ since, by convention, (i, j, k) is a basis of V^3 .

$$\pi: \begin{cases} x = 1 + t \cdot 1 + s \cdot 0 \\ y = 2 + t \cdot 0 + s \cdot 1 \\ z = 1 + t \cdot 0 + s \cdot 0 \end{cases} \Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad t, s \in \mathbb{R}$$

$\downarrow$
 A

d) π contains $M(1,7,1)$ and is $\parallel$ to the coordinate plane Oyz .

• a basis for $D(Oyz)$ is (j, k) .

$\downarrow \downarrow$
 $j \quad k$

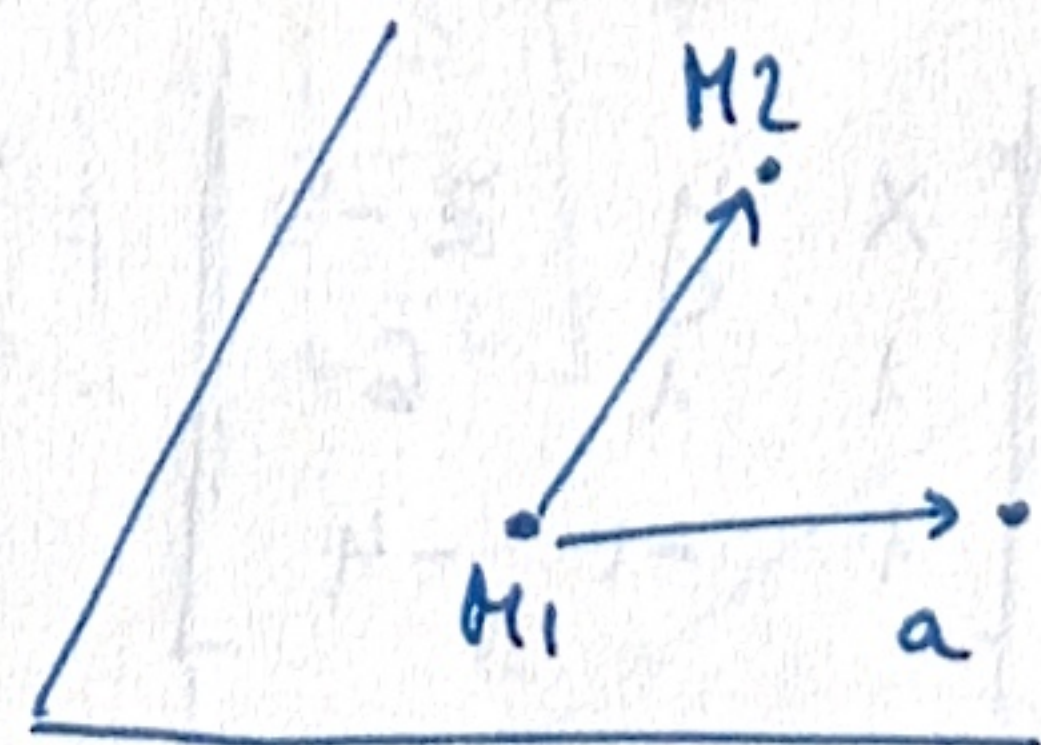
$$j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\pi: \begin{cases} x = 1 \\ y = 7 + t \\ z = 1 + s \end{cases}, t, s \in \mathbb{R}. \quad \Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 1 \end{bmatrix} + t \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad t, s \in \mathbb{R}.$$

e) π contains $M_1(5,3,4)$, $M_2(1,0,1)$ and is $\parallel$ to $a(1,3,-3)$.

$\overrightarrow{M_2 M_1} (5-1, 3-0, 4-1) = \overrightarrow{M_2 M_1} (4, 3, 3)$ is a vector represented with points in $\pi \Rightarrow \pi \parallel \overrightarrow{M_2 M_1}$.

vectors $\overrightarrow{M_2 M_1}$ and a are lin. ind. vectors $\Rightarrow$
 $\Rightarrow$ they form a basis of $D(\pi)$.



$$\pi: \begin{cases} x = 5 + 4t + s \\ y = 3 + 3t + 3s \\ z = 4 + 3t + (-3)s \end{cases}, t, s \in \mathbb{R}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $M_1 \quad \overrightarrow{M_2 M_1} \quad a$

$$\Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix} \cdot t + \begin{bmatrix} 1 \\ 3 \\ -3 \end{bmatrix} \cdot s, \quad t, s \in \mathbb{R}.$$

2.4. Det. Cartesian eq. for the plane π :

a) $\pi: x=2+3u-4v, y=4-v, z=2+3u, u,v \in \mathbb{R}.$

$$\pi: \begin{cases} x=2+3u-4v \\ y=4-v \\ z=2+3u \end{cases}, u,v \in \mathbb{R}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} + 3 \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot u - \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} v, u,v \in \mathbb{R}$$

$$\Rightarrow \begin{bmatrix} x-2 \\ y-4 \\ z-2 \end{bmatrix} = \underset{a}{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}} \cdot 3u - \underset{b}{\begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}} \cdot v, u,v \in \mathbb{R}.$$

$$\pi: \begin{vmatrix} x-2 & y-4 & z-2 \\ a & & \end{vmatrix} = 0$$

$$a \rightarrow \begin{vmatrix} 1 & 0 & 1 \\ 4 & 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(-1) - (y-4)(-4) + (z-2) = 0$$

$$\Rightarrow \pi: -x + 4y + z - 16 = 0.$$

b) $\pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix} \cdot u + \begin{bmatrix} 1 \\ -1 \\ -4 \end{bmatrix} v, u,v \in \mathbb{R}.$

$$\Rightarrow \pi: \begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = 0$$

$$\Rightarrow \pi: 2x + 10y - 2 \cdot 2 + 10 = 0 \quad |:2$$

$$\Rightarrow \pi: x + 5y - z + 5 = 0$$

Compute determinant M1: $x \cdot \underbrace{\begin{vmatrix} 1 & 6 \\ -1 & -4 \end{vmatrix}}_2 - y \cdot \underbrace{\begin{vmatrix} 1 & 6 \\ 1 & -4 \end{vmatrix}}_{-10} + (z-5) \cdot \underbrace{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}_{-2} = 0$

Compute determinant M2:

$$\begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = \underline{-4x} + \underline{5z} - 2 + \underline{6y} + \underline{4y} + \underline{6x} + 5 - z$$

$$= 2x + 10y + 10 - 2z$$

$$\begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \end{vmatrix}$$

2.5. Det. parametric eq. for the plane π :

a) $\pi: 3x - 6y + z = 0$

Choose x and y as parameters : $\begin{cases} x = x \\ y = y \\ z = -3x + 6y \end{cases}$

$$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix} \quad t, s \in \mathbb{R}.$$

we "relabel" x and y .

b) $\pi: 2x - y - z - 3 = 0$.

Choose $x=t$ and $z=s$ as parameters : $\begin{cases} x = t \\ y = 2x - z - 3 = 2t - s - 3 \\ z = s \end{cases}$

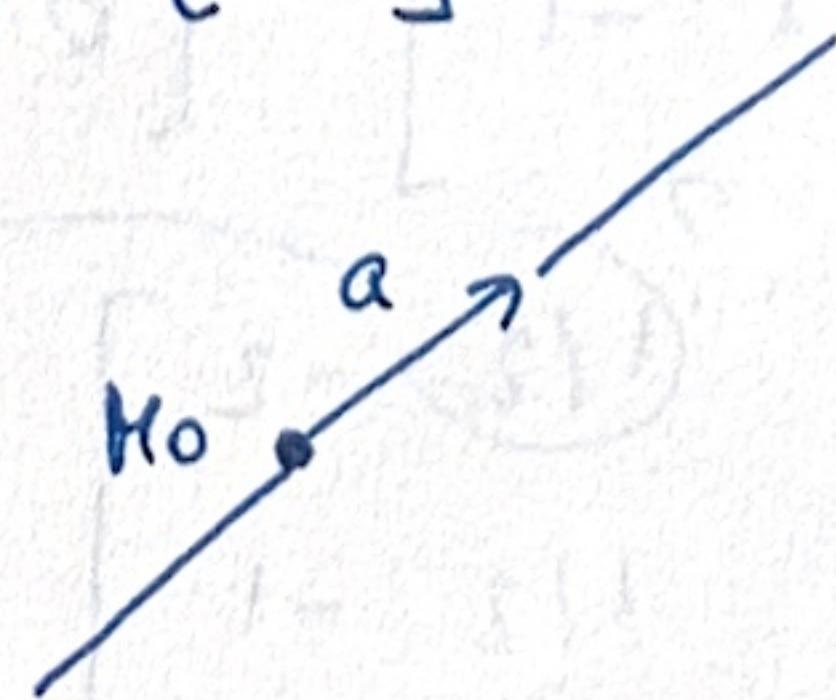
$$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

↓
ct.
(point)

2.6. Det. parametric eq. for the line l :

a) l contains $M_0(2, 0, 3)$ and is $\parallel$ to $a(3, -2, -2)$.

$$l: \begin{cases} x = 2 + 3t \\ y = 0 - 2t \\ z = 3 - 2t \end{cases}, t \in \mathbb{R} \quad (\Rightarrow) \quad l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix} \cdot t$$



b) l contains $A(1, 2, 3)$ and is $\parallel$ to Oz axis.

$\hookrightarrow k$ is a direction vector, $k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

$$l: \begin{cases} x = 1 \\ y = 2 \\ z = 3 + 1 \cdot t \end{cases}, t \in \mathbb{R} \quad (\Rightarrow) \quad l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \cdot t$$

c) l contains $M_1(1, 2, 3)$ and $M_2(4, 4, 4)$.

• a direction vector is $\overrightarrow{M_1M_2}$ or $\overrightarrow{M_2M_1}$

• a point on l is M_1 or M_2 .

2.7. Cartesian eq. for the lines l in 2.6.

$$a) l: \begin{cases} x=2+3t \Rightarrow t = \frac{x-2}{3} \\ y=-2t \Rightarrow t = -\frac{y}{2} \\ z=3-2t \Rightarrow t = \frac{z-3}{-2} \end{cases} \Rightarrow \frac{x-2}{3} = \frac{y}{-2} = \frac{z-3}{-2}$$

Choose 2 equations: $l: \begin{cases} \frac{x-2}{3} = \frac{y}{-2} \\ \frac{y}{-2} = \frac{z-3}{-2} \end{cases} \Leftrightarrow l: \begin{cases} -2x+4=3y \\ -2y=-2z+6 \end{cases}$

$$\Leftrightarrow l: \begin{cases} 2x+3y-4=0 \\ 2y-2z+6=0 \end{cases}$$

$$b) l: \begin{cases} x=1 \\ y=2 \\ z=3+t \rightarrow z \text{ is arbitrary} \end{cases} \Rightarrow l: \begin{cases} x=1 \\ y=2 \end{cases} \Leftrightarrow \bar{l}: \begin{cases} x-1=0 \\ y-2=0 \end{cases}$$

?!

2.8. Parametric eq. for the line contained in the planes
 $x+y+2z-3=0$ and $x-y+z-1=0$.

$$l: \begin{cases} x+y+2z-3=0 \\ x-y+z-1=0 \end{cases} (*)$$

$$\begin{matrix} 2 - \frac{1}{2} \\ -3 + 1 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & -3 \\ 1 & -1 & 1 & -1 \end{bmatrix} \xrightarrow{L_2 - L_1 \rightarrow L_2} \begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & -2 & -1 & 2 \end{bmatrix} \xrightarrow{-\frac{1}{2} \cdot L_2 \rightarrow L_2} \begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & 1 & 1/2 & -1 \end{bmatrix} \xrightarrow{L_1 - L_2 \rightarrow L_1} \begin{bmatrix} 1 & 0 & 3/2 & -2 \\ 0 & 1 & 1/2 & -1 \end{bmatrix}$$

$$(*) \Leftrightarrow \begin{cases} x + \frac{3z}{2} - 2 = 0 \\ y + \frac{z}{2} - 1 = 0 \end{cases}$$

$\rightarrow$ why?

if we choose the parameter $t = z/2$, then:

$$\begin{cases} x = 2 - \frac{3z}{2} \\ y = 1 - \frac{z}{2} \end{cases} \Leftrightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}$$

2?

2.9. Cartesian eq. for the plane containing point Q and line l.

a) $Q = (3, 3, 1)$, $l: x = 2 + 3t, y = 5 + t, z = 1 + 7t, t \in \mathbb{R}$.

$$Q = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}, l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}}_A + t \cdot \underbrace{\begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}}_N, t \in \mathbb{R}$$

$\Rightarrow$ the plane contains points Q and A
 $\parallel N$.

$$\vec{QA} = A - Q = (2, 5, 1) - (3, 3, 1) = (-1, 2, 0)$$

$$\Rightarrow \vec{QA} (-1, 2, 0). \Rightarrow \pi \parallel \vec{QA}.$$

$\vec{QA}, N$ are lin. ind. vectors $\Rightarrow$ they form a basis of $D(\pi)$.

$$\pi: \begin{cases} x = -1 + (-1) \cdot t + 3 \cdot s \\ y = 2 + 2 \cdot t + 1 \cdot s \\ z = 0 + 0 \cdot t + 7 \cdot s \end{cases} \quad t, s \in \mathbb{R} \quad ?$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $Q \quad \quad \vec{QA} \quad \quad N$

$$\pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \cdot t + \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix} \cdot s \quad t, s \in \mathbb{R}.$$